Ch 13: Statistics

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. Find the class mark of the class 25-35. Also state the formula for mean using assumed mean method.

Q2. Write the formula for mean using step-deviation method. State what each symbol represents.

Q3. Write the formula for median of grouped data. State what each symbol represents.

Q4. Write the formula for mode of grouped data. State what each symbol represents.

Q5. State the empirical relationship between mean, median, and mode.

Q6. If mean = 25.5 and median = 26, find the mode using the empirical formula.

Q7. If mean = 18 and mode = 14, find the median using the empirical formula.

Q8. Find the modal class and mode of the following data:

Class	Frequency
10-20	4
20-30	8
30-40	12
40-50	7
50-60	4

Q9. Find the median class of the following distribution:

Class	Frequency
0-10	5
10-20	8
20-30	12
30-40	10
40-50	5

Q10. For a less-than ogive, what values are plotted on the x-axis and y-axis? How is it different from a more-than ogive?

Q11. What is the x-coordinate of the intersection point of the less-than and more-than ogives? Why?

Q12. Describe in brief the procedure to find the median from an ogive.

Q13. The mean of five numbers is 18. If one number is excluded, the mean becomes 16. Find the excluded number.

Q14. If the mean of first n natural numbers is 15, find n.

Q15. The mean of 30 numbers is 18. If three numbers 20, 25, and 30 are removed, find the new mean.

Q16. A data has mean 35 and median 33. Find the mode. Is this data positively or negatively skewed?

Q17. The following distribution has mean 50. Find the missing frequency f. (Given: total frequency = 100)

Class	Frequency
0-20	17
20-40	f
40-60	32
60-80	24
80-100	19

Q18. In a grouped frequency distribution, class size h = 10 and assumed mean A = 25. If sum(fi*ui) = 30 and sum(fi) = 50, find the mean.

Q19. If sum(fi) = 100 and sum(fi*xi) = 4750, find the mean.

Q20. Find the mean of the first 10 multiples of 3.

Q21. The median of the following data is 24. Find x.

Class	Frequency
0-10	4
10-20	8
20-30	x
30-40	7
40-50	5

Q22. Convert the following more-than frequency distribution to a simple frequency table:

More than (Lower Limit)	Cumulative Frequency
10	48
20	40
30	25
40	12
50	5

Q23. Can mean, median, and mode all be equal for a distribution? Give an example.

Q24. Give one situation each where (i) mean is the best measure and (ii) mode is the best measure of central tendency.

Q25. The mean of 20 observations is 15. On checking, observations recorded as 12 and 18 were actually 21 and 9. Find the correct mean.

Q26. Construct a less-than cumulative frequency table for the following data:

Upper Class Limit	Cumulative Frequency (Less than)
10	5
20	13
30	23
40	29
50	32

Q27. Construct a more-than cumulative frequency table for the data in Q26.

Lower Class Limit	Cumulative Frequency (More than)
0	32
10	27
20	19
30	9
40	3

Q28. If sum(fi) = 50, A = 25, h = 10, and sum(fi*ui) = 30, find the mean by step-deviation method.

Q29. Two distributions have the same mean. Does that mean they are identical? Explain briefly.

Q30. Find the mode of the following data:

Class	Frequency
0-20	6
20-40	8
40-60	15
60-80	9
80-100	2

Q31. In a grouped frequency distribution, the class intervals are 10-20, 20-30, 30-40 with frequencies 5, 18, 7. Which is the modal class?

Q32. The mean of the following frequency distribution is 62.8. Find the missing frequency f. (Given: total frequency = 50)

Class	Frequency
0-20	5
20-40	8
40-60	f
60-80	12
80-100	7

SECTION B - 3 MARKS

Q1. Find the mean of the following distribution using the direct method: (PYQ 2020)

Class	Frequency
0-10	8
10-20	16
20-30	36
30-40	34
40-50	6

Q2. Find the mean of the following using the assumed mean method (A = 25):

Class	Frequency
0-10	12
10-20	14
20-30	8
30-40	6
40-50	10

Q3. Find the mean of the following distribution using the step-deviation method: (PYQ 2019)

Class	Frequency
10-25	2
25-40	3
40-55	7
55-70	6
70-85	6
85-100	6

Q4. Find the mean daily wages of workers in a factory from the following data: (PYQ 2020)

Wages (Rs.)	Workers
100-120	10
120-140	15
140-160	20
160-180	22
180-200	18

Q5. Find the mean of the following frequency distribution using the step-deviation method:

Class	Frequency
0-100	2
100-200	5
200-300	14
300-400	15
400-500	4

Q6. Find the missing frequency p in the following distribution if the mean is 28.5. (Given: total frequency = 60) (PYQ 2019)

Class	Frequency
10-20	5
20-30	p
30-40	20
40-50	15
50-60	7
60-70	5

Q7. Find the missing frequencies p and q if the mean is 42 and total frequency = 70: (PYQ 2020)

Class	Frequency
0-10	7
10-20	10
20-30	p

30-40	13
40-50	q
50-60	10
60-70	14

Q8. Find the median of the following data: (PYQ 2019)

Class	Frequency
0-10	5
10-20	8
20-30	20
30-40	15
40-50	7

Q9. Find the median of the following distribution: (PYQ 2020)

Class	Frequency
100-120	12
120-140	14
140-160	8
160-180	6
180-200	10

Q10. Find the median of the following data: (PYQ NCERT)

Class	Frequency
500-600	36
600-700	32
700-800	32
800-900	20
900-1000	30

Q11. Compute the median for the following less-than type cumulative frequency distribution:

Upper Limit (Less than)	Cumulative Frequency
10	5
20	13
30	25
40	32
50	35

Q12. Find the median from the following more-than type distribution:

Lower Limit (More than)	Cumulative Frequency
150	80
160	77
170	60
180	30
190	8

Q13. Find the mode of the following distribution: (PYQ 2019)

Class	Frequency
0-20	6
20-40	8
40-60	15
60-80	9
80-100	2

Q14. Find the mode of the following data: (PYQ 2020)

Class	Frequency
25-30	25
30-35	34
35-40	50
40-45	42
45-50	38
50-55	14

Q15. Find the mode of the following distribution and verify using empirical formula if median = 35:

Class	Frequency
10-20	5
20-30	9
30-40	12
40-50	8
50-60	4
60-70	2

Q16. Find the missing frequency if the median of the following data is 24:

Class	Frequency
0-10	4
10-20	8
20-30	x
30-40	7
40-50	5

Q17. Find the missing frequencies p and q if the median is 33 and total frequency is 60: (PYQ 2019)

Class	Frequency
10-20	5
20-30	p
30-40	20
40-50	15
50-60	q
60-70	5

Q18. The annual rainfall data of a town for 70 years is as follows. Compute the median rainfall:

Rainfall (cm)	Years
25-35	5

35-45	13
45-55	20
55-65	19
65-75	8
75-85	5

Q19. Find the mean using step-deviation method and the mode. Verify using empirical formula: (PYQ 2022)

Class	Frequency
0-10	3
10-20	9
20-30	15
30-40	30
40-50	18

Q20. Find the mean height using assumed mean method:

Height (cm)	Students
150-155	5
155-160	11
160-165	14
165-170	16
170-175	4

Q21. Find the mean, median, and check if the empirical formula holds approximately:

Weight (kg)	Students
40-44	4
44-48	12
48-52	15
52-56	6
56-60	3

Q22. Find the missing frequencies a and b if mean = 53 and total frequency = 100:

Class	Frequency
0-20	15
20-40	a
40-60	21
60-80	b
80-100	17

Q23. Find the electricity bill mean using step-deviation method:

Bills (Rs.)	Houses
200-300	3
300-400	5
400-500	10
500-600	4
600-700	3

Q24. Find the mean daily income of 50 workers using assumed mean method:

Daily Income (Rs.)	Workers
100-120	12
120-140	14
140-160	8
160-180	6
180-200	10

Q25. Find the mode of the following and verify using the mean and median:

Class	Frequency
10-20	4
20-30	8
30-40	11
40-50	15
50-60	12

60-70	6

Q26. Find the marks mean, median, and mode: (PYQ 2023)

Marks	Students
0-20	6
20-40	17
40-60	15
60-80	16
80-100	26

Q27. Find the median of the following grouped data using the formula:

Age (yrs)	Frequency
5-15	6
15-25	11
25-35	21
35-45	23
45-55	14
55-65	5

Q28. Find the mode of the following distribution. Use the empirical formula to find the mean if median = 40:

Class	Frequency
0-10	5
10-20	8
20-30	14
30-40	10
40-50	7
50-60	6

Q29. Find the mean of the following using step-deviation method. Also find the modal class:

Class	Frequency

0-20	15
20-40	18
40-60	22
60-80	25
80-100	10

Q30. The marks in Science of 80 students are given. Find the mode and median:

Marks	Students
0-10	3
10-20	5
20-30	16
30-40	12
40-50	13
50-60	20
60-70	5
70-80	4
80-90	1
90-100	1

SECTION C - 5 MARKS

[Less Than Type Ogive]

Q1. Draw a less-than type ogive for the following data. Hence obtain the median from the graph and verify using the median formula: (PYQ 2020)

Upper Limit (Less than)	Cumulative Frequency
10	5
20	12
30	22
40	35
50	45
60	53

Q2. The following data gives the observed lifetimes of 225 electrical components. Draw a less-than type ogive and find the median graphically: (PYQ NCERT)

Upper Limit (Less than)	Cumulative Frequency
20	10
40	45
60	97
80	158
100	196
120	225

Q3. The following table shows marks obtained by 100 students. Draw a less-than type ogive. Find: (i) the median graphically (ii) the number of students who scored more than 75 marks: (PYQ 2022)

Upper Limit (Less than)	Cumulative Frequency
20	5
40	15
60	45
80	85

100	100

Q4. Draw a less-than type ogive for the following distribution. Locate the median on the graph and verify using the formula: (PYQ)

Upper Limit (Less than)	Cumulative Frequency
10	8
20	25
30	50
40	70
50	85
60	92
70	100

[More Than Type Ogive]

Q5. Draw a more-than type ogive for the following data and find the median graphically. Verify using the formula: (PYQ)

Lower Limit (More than)	Cumulative Frequency
0	100
10	92
20	75
30	50
40	25
50	8

Q6. The following data gives the production yield of 100 farms. Draw a more-than type ogive and find the median: (PYQ NCERT)

Lower Limit (More than)	Cumulative Frequency
50	100
55	98
60	90
65	78

70	54
75	16

Q7. Draw a more-than type ogive for the following distribution. Locate the median graphically and verify: (PYQ)

Lower Limit (More than)	Cumulative Frequency
10	80
20	77
30	60
40	46
50	25
60	10
70	3

Q8. The annual profits earned by 30 shops are given. Draw a more-than type ogive and find the median from the graph: (PYQ 2019)

Lower Limit (More than)	Cumulative Frequency
5	30
10	27
15	13
20	7
25	3
30	1
35	0

Q9. The following distribution gives the monthly consumption of electricity of 68 consumers. Find mean, median, and mode: (PYQ NCERT)

Units	Frequency
65-85	4
85-105	5
105-125	13
125-145	20

145-165	14
165-185	8
185-205	4

Q10. The following frequency distribution shows the number of runs scored by some batsmen. Find mean (step-deviation method), median, and mode: (PYQ 2022)

Runs	Batsmen
2000-4000	9
4000-6000	8
6000-8000	10
8000-10000	2
10000-12000	1

Q11. Find the missing frequencies in the following distribution if mean = 57 and total frequency = 70: (PYQ 2020)

Class	Frequency
0-20	10
20-40	f1
40-60	25
60-80	30
80-100	f2
100-120	10

Q12. Find the missing frequencies p and q if the median = 46 and total frequency = 230: (PYQ 2019)

Class	Frequency
10-20	12
20-30	30
30-40	p
40-50	65
50-60	q

60-70	25
70-80	18

Q13. Find mean, median, and mode. Draw both ogives and verify the median graphically:

Height (cm)	Students
140-145	8
145-150	7
150-155	18
155-160	12
160-165	8

Q14. Find the daily pocket allowance mean (assumed mean method) and the missing frequency. Given: Mean = 18. (PYQ NCERT)

Allowance (Rs.)	Children
11-13	7
13-15	6
15-17	9
17-19	13
19-21	f
21-23	5
23-25	4

Q15. Case Study: A school conducted a survey of 100 students scores in a test. Find: (i) mean using step-deviation method (ii) modal class and mode (iii) median class and median (iv) verify: 3 Median = Mode + 2 Mean approximately. (PYQ 2023)

Marks	Students
0-20	7
20-40	13
40-60	35
60-80	30

80-100	15

Q16. Find the mean wage (assumed mean method), modal wage, and draw a less-than ogive to find the median wage:

Wages (Rs.)	Workers
100-150	20
150-200	22
200-250	28
250-300	24
300-350	16
350-400	10

Q17. Find the mean, mode, and median of the following data and comment on the skewness of the distribution:

Class	Frequency
10-20	8
20-30	11
30-40	17
40-50	15
50-60	10
60-70	9

Q18. Find mean, median, and mode. Verify the empirical relation. Draw less-than ogive and find the median graphically: (PYQ 2024)

Class	Frequency
0-10	6
10-20	10
20-30	14
30-40	8
40-50	2

Q19. A survey regarding the heights (in cm) of 51 girls of Class X was conducted. Find the mean, median, and mode: (PYQ NCERT)

Height (cm)	Girls
120-130	2
130-140	8
140-150	12
150-160	20
160-170	9

Q20. Find the median weight. Draw both ogives and verify: (PYQ NCERT)

Weight (kg)	Students
40-45	2
45-50	3
50-55	8
55-60	6
60-65	6
65-70	3
70-75	2

Q21. Find mean, mode, median, and draw less-than ogive. Verify median graphically: (PYQ NCERT)

Age (yrs)	Patients
5-15	6
15-25	11
25-35	21
35-45	23
45-55	14
55-65	5

Q22. Find the values of p and q if mean = 18 and total frequency = 60. Also find the median and mode: (PYQ 2020)

Variate	Frequency

10	p
15	15
20	10
25	q
30	2

Q23. Following is the age distribution of a group of students. Draw both ogives and find the median age. Verify using the formula: (PYQ 2020)

Age	Students
4-5	90
5-6	150
6-7	75
7-8	225
8-9	210
9-10	105

Q24. Find the mean of the following distribution using all three methods and verify they give the same result:

Class	Frequency
0-20	5
20-40	10
40-60	20
60-80	10
80-100	5

Q25. Find the monthly wages mean, median, and mode of the workers. Draw a more-than ogive and verify the median graphically:

Wages (Rs.)	Workers
100-150	20
150-200	22
200-250	28
250-300	24

300-350	16
350-400	10